FRI
BINGHAMTON UNIVERSITY
STATE UNIVERSITY OF NEW YORK

Allen Fertidos

Mert Kanibir

Jordan Lee

Eldho Thomas

Jonathan Zhang

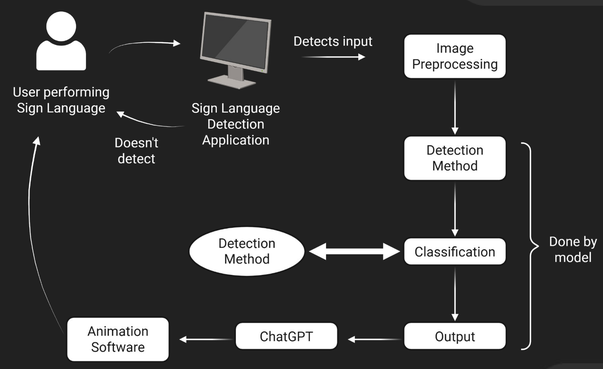
# YANA: Your ASL Native Assistant

What is the Introduction?

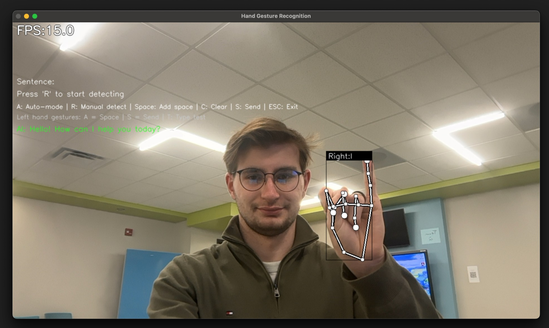
- Why is sign language important?
  - Gives a voice to many mute / deaf individuals
  - Creates more accessible and inclusive communication.
- Sign language is essential for communication within the deaf and hard-of-hearing community
- Many AI assistants like Siri or Alexa use voice as the main mode of communication, excluding millions of people who use sign language

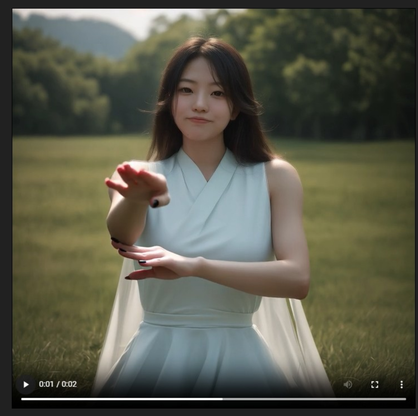
Can you explain the Methods?

- Camera captures live ASL gestures
- Input is preprocessed to enhance image quality
- CNN model analyzes and recognizes hand signs
- Detected signs are sent into a ChatGPT API and displays text output
- Library of animated videos of ASL signs is created with Wan 2.2 and OpenPose detection
- ChatGPT's output is reconstructed using selected videos and displayed to the user




What are the Results?




- Achieved moderate to high accuracy in sign recognition
- Successfully translated common ASL gestures
- Performance affected by lighting and complex movements

- Video output of a generated character signing the phrase "Good Afternoon" back

What is the Conclusion?

YANA demonstrates how AI can enhance communication accessibility in digital settings for the deaf community. While current performance is promising, future improvements in model training and gesture diversity will further increase accuracy and reliability.

- YANA can have great potential in applications in educational, health, and customer services

What other questions do you have about **YANA**?

## References

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YANA
Free

Demo

Ask anything

YANA can make mistakes. Check important info.